Trilobite generator — changes of 20 September 2026

print joint, print fill, reconciliation with the 16:50 tree

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Summary

This delta sits on top of the tree zipped at 16:50 on 20 September (`trilobite_generator_code_2026-09-20_1650.z` schema 6.1, AST `safe_expr`, `manifold3d==3.0.1`, references re-frozen, PREREG resolved). It adds nothing to the measured geometry or the instrument's ruler except one pre-registered rule that had been dropped from the code. Everything else is *print-only* and lives behind two optional request fields on `/api/build`.

file	status	what
`instrument.py`	modified	PREREG §3 overlap term restored in `classify()`
`web/app.py`	modified	`"joint": "flexi"` and `"fill": <mm>` on `/api/`
`printjoint2.py`	new	the print joint (parametrized Flexi #3839472), wh
`printfill.py`	new	print-only shell thickening for the pin build
`tests/test_printjoint2.py`	new	2 tests, ∼4 s
`scripts/printjoint2_study.py`, `_sliders.py`	new	joint study; one-slider-at-a-time study
`docs/BLOCK1_DECISIONS.md`, `block1_*.json`	new	Block 1 gate and spine evidence (for the record)
`CHANGES_2026-09-13.md`	appended	the 20 Sep entries

1 The instrument: one change

`classify()` implemented "closed with total $< 150°$ ⇒ discoidal". PREREG §3 says discoidal requires *both* the short curl *and* pygidium–cephalon overlap $s_{\text{tail}} \geq$ `DISC_OVERLAP` (resolved: 0.5). Restored verbatim; short-curl closures without the overlap now fall through to the s_{tail} rules. The constants (`DISC_DEG`, `DISC_OVERLAP`, `CLASS_TOL`) are now written into every result. Agnostida stays discoidal ($s_{\text{tail}} = 1.56$); no frozen reference changes. This closes the last "still open" item in the changelog.

2 The print joint (`printjoint2.py`)

Two Thingiverse print-in-place trilobites were taken apart in code (#7112261: 31 bodies, T-tenons in loose pockets; #3839472: 8 bodies, one knob-in-pocket joint per pair). The second is the one adopted, parametrized on our segment geometry:

- every segment is a *solid wedge on a flat base* (no hollow shell, no nesting, no supports);

- one joint per pair at the midline: a short *barrel* (axis across the body) on a neck, reaching back from segment k into a box pocket inside segment $k{+}1$ through a slot in its front face. A barrel, not a disc or sphere, because its fore–aft extent does not change with flexion: pull play equals the axial gap (0.10–0.25 mm) instead of the 0.6 mm the Thingiverse disc needs;
- V-cut faces below the pivot set the stop at `maxAngle`; measured stop $=$`maxAngle`$\pm 2^\circ$ on every slider case and every preset;
- three clearances by print direction (axial 0.20, vertical 0.30, lateral 0.25 mm), a 1.4 mm lip, a 45° bed chamfer;
- the pivot is an absolute height (6–10 mm above the bed), not a fraction of relief; the joint scales itself down when the pitch is within 1 mm of what it needs and refuses below 4.5 mm pitch.

Whole animals: `print_head`, `print_segment`, `print_tail`, `print_animal`; posed with `printjoint2.transforms_d` (the print pivot sits ≈ 2.4 mm behind the instrument's joint plane — state it, do not pose print parts with the instrument's transforms). All ten presets: every part one watertight body, no contact at the stop, including head–tail. Retention: the barrel's 0.65 mm side shoulders and 0.50 mm top/bottom shoulders bear on the lip; overlap volume rises steadily past the 0.2 mm play.

Known limits. A ventral curl on solid segments costs base width: at `maxAngle` 36° the V cut leaves 43 % of the pitch on the bed (72 % at 18°). The presets carry `maxAngle` 18, so printed presets make a C, not a ball; the export should take its stop from the instrument's reading for that animal.

3 Print fill (`printfill.py`)

Printed pin-build animals are the 2 mm measurement shell and are too thin to hold. `thicken()` adds the layer between the dorsal surface and the same surface shifted down `fill` mm. Hinge bands (barrels, bores, knuckles), the flap that rides over the next segment, and the pleural tips (depth $\leq$ `marginHeight`$\cdot$`relief`-1.5, never below $z = 0.6$) are kept clear. Default animal at 3 mm: 26 g $\rightarrow$ 32 g, single bodies, no rest contact, clean to its 18° stop. Three mm is the ceiling for the pin build (the shells nest at 2.3 mm). The measured geometry and the parameter hash are untouched: the sheet still describes the 2 mm animal.

4 Web

`/api/build` body may carry `"joint": "pin"|"flexi"` (default pin) and `"fill": <mm>` (pin only). Cache keys get `-flexi` / `-fill3` suffixes so measured and print builds never collide; the manifest records `joint`, `fill_mm` and, for flexi, a `print_joint` block (pivot, offset, scale, gaps). `/api/stl` poses flexi builds with the print pivot. `index.html` is unchanged — no toggle yet; one checkbox beside *Download STL* is all it needs.

5 Evidence carried in `docs/`

`block1_gate.json`: pleural facet + ventral doublure move e@stop by -0.08, -0.16, -1.09, 0.00, 0.00° on the five anchors, no class flips — the reason schema 6.2's keys were *not* adopted. `block1_spines.json`: the four pilot anatomy readings are identical with every spine family zeroed. `BLOCK1_DECISIONS.md`: the reasoning.

6 Open items surfaced today (not fixed here)

1. A stalked eye (`eyeStalk`> 0) comes out as three bodies in the *tracked* `parts.cephalon` (drum floating, stalk missing). Reproduced on asaphida with `eyeStalk`=2.0. Generator bug from 6.1; no preset uses a stalk, so references are unaffected. Block 2.
2. Seven of ten presets have `eyeSolid`=0 and print blind; harpetida's eye is below the lens threshold.
3. The 09-16 print fused because of the nested-shell design at 0.3 mm, not the printer; `print_validity` still checks feature size, not gap, and should gain a gap check.
4. Renders made 13–20 Sep before the `manifold3d` pin ran on 3.5.3 and carry its cylinder-seam spike; the joint numbers were re-verified on 3.0.1 (unchanged).

7 How to commit

```
# from the repo root at the 16:50 tree (schema 6.1, manifold3d==3.0.1)
git checkout -b print-joint-and-fill
unzip -o trilobite_delta_2026-09-20.zip -d . # 14 files: 2 modified, 12 new
pip install -r requirements.txt # manifold3d==3.0.1 is required
python -m pytest tests/test_printjoint2.py tests/test_schema.py -q # ~2.5 min

git add instrument.py web/app.py printjoint2.py printfill.py \
       tests/test_printjoint2.py scripts/printjoint2_study.py scripts/printjoint2_sliders
           .py \
       docs/ CHANGES_2026-09-13.md
git commit -m "classify: restore PREREG s3 overlap term; print joint (flexi) + print fill
    behind /api/build fields

- instrument.classify: discoidal needs total<150 AND s_tail>=DISC_OVERLAP (0.5);
    constants in results.
  No reference changes.
- printjoint2.py: parametrized Flexi #3839472 joint on our segments (solid wedges, barrel
    -in-pocket,
  V stop at maxAngle, direction-split gaps, self-scaling below pitch 7.4, refuses <4.5).
    Whole-animal
  builders and print-pivot transforms. All presets: single bodies, no contact at the stop
    .
- printfill.py: print-only shell thickening for the pin build (hinge bands and nesting
    flap kept clear).
- web/app.py: /api/build accepts joint=pin|flexi and fill=<mm>; cache keys suffixed; STL
    posing.
- tests/test_printjoint2.py; docs/: Block 1 gate + spine evidence.
Measured geometry, schema (6.1), hashes and frozen references unchanged."
```

Then, separately and only if you want it: the `index.html` checkbox (`joint` + `fill` in the build request) and the stalked-eye fix in `parts.cephalon`, each as its own commit so the print work stays reviewable on its own.